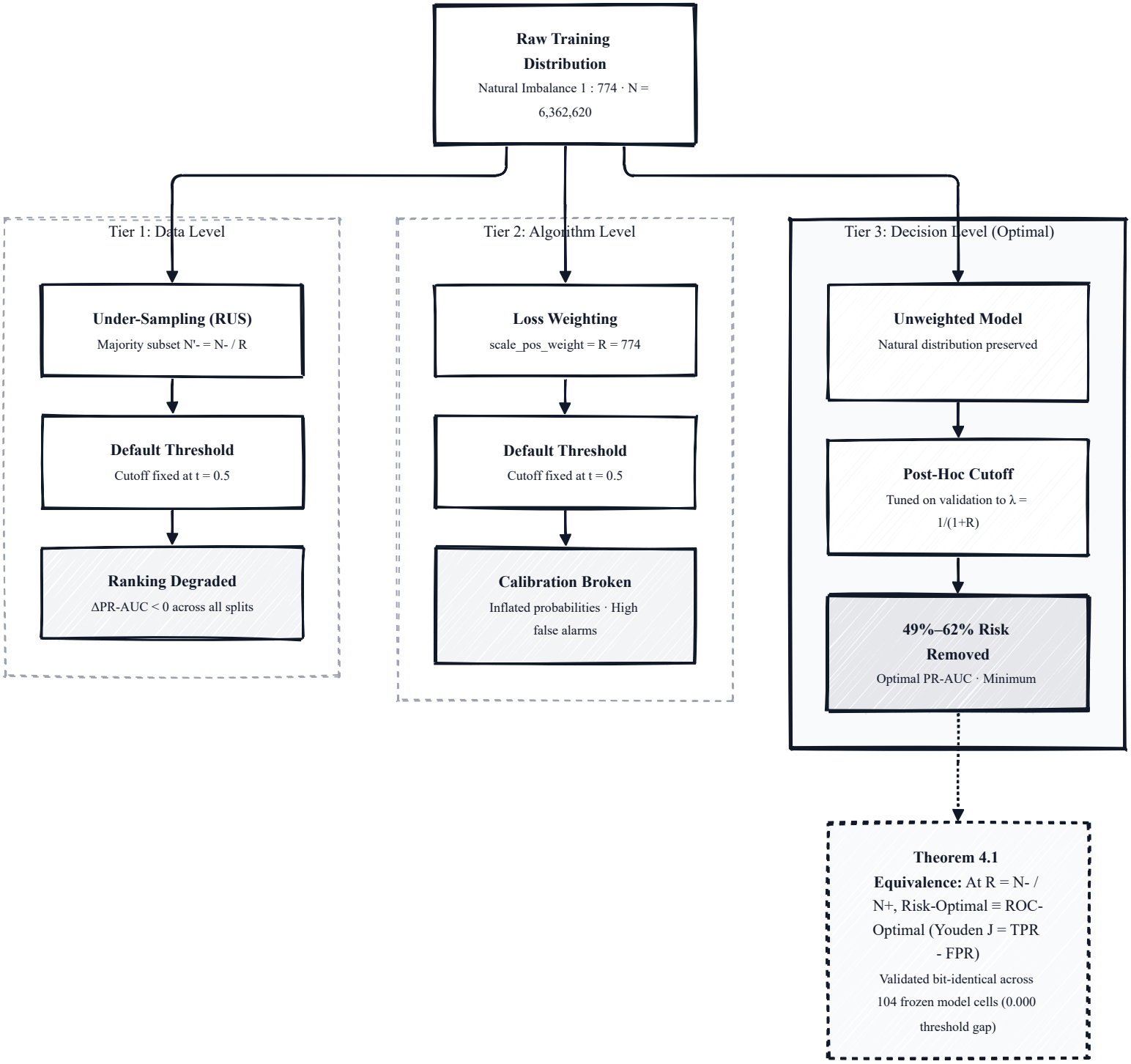

Raw Training Distribution
Natural Imbalance 1 : 774 · N = 6,362,620
Tier 1: Data Level
Under-Sampling (RUS)
Majority subset N'- = N- / R
Default Threshold
Cutoff fixed at t = 0.5
Ranking Degraded
ΔPR-AUC < 0 across all splits
Tier 2: Algorithm Level
Loss Weighting
scale_pos_weight = R = 774
Default Threshold
Cutoff fixed at t = 0.5
Calibration Broken
Inflated probabilities · High false alarms
Tier 3: Decision Level (Optimal)
Unweighted Model
Natural distribution preserved
Post-Hoc Cutoff
Tuned on validation to λ = 1/(1+R)
49%–62% Risk Removed
Optimal PR-AUC · Minimum
Theorem 4.1
Equivalence: At R = N- / N+, Risk-Optimal ≡ ROC-Optimal (Youden J = TPR - FPR)
Validated bit-identical across 104 frozen model cells (0.000 threshold gap)